HCF and LCM: prime factors

In this sheet you will…

- break a number into prime factors with a division ladder and with a factor tree
- find the highest common factor (HCF) of two numbers
- find the lowest common multiple (LCM) of two numbers
- know what each one means and what it is used for

Key words

prime number
factor · multiple
prime factor
factorisation
HCF · LCM

This sheet is a reminder of what you studied in S1. **Revise this very carefully, especially if you have trouble with this topic.**

1 Prime numbers and prime factors

Prime number

A number with exactly **two** factors: 1 and itself. 2, 3, 5, 7, 11, 13, … (1 is not prime.)

Prime factors

The prime numbers that multiply to give the number. Every number above 1 has **only one** way to be written as a product of primes.

2 Two methods to find the prime factors

Method A: division ladder

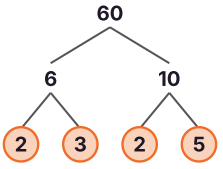
Divide by the smallest prime that fits. Keep going until you reach 1.

2	60
2	30
3	15
5	5
	1

$60 = 2 \times 2 \times 3 \times 5 = 2^2 \times 3 \times 5$

Method B: factor tree

Split into any two factors. Keep splitting until every branch ends in a prime (circled).



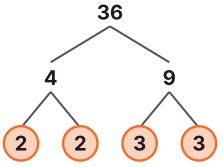
$60 = 2 \times 2 \times 3 \times 5 = 2^2 \times 3 \times 5$

Ladder again

2	36
2	18
3	9
3	3
	1

$36 = 2 \times 2 \times 3 \times 3 = 2^2 \times 3^2$

Tree again



$36 = 2 \times 2 \times 3 \times 3 = 2^2 \times 3^2$

Tip: 2^2 is a short way to write 2×2. You can write the factorisation either way. **Different trees give the same primes:** 36 can start as 4×9 or as 2×18, and you always end with $2 \times 2 \times 3 \times 3$.

HCF and LCM

3 Highest Common Factor (HCF)

What does HCF mean?

The **biggest number that divides exactly into both numbers**. **Used for:** simplifying fractions.

Worked example 2: find the HCF of 36 and 60

Step 1 Break both numbers down (ladder or tree):

$36 = 2 \times 2 \times 3 \times 3 = 2^2 \times 3^2$

$60 = 2 \times 2 \times 3 \times 5 = 2^2 \times 3 \times 5$

Step 2 Mark the primes that **both** numbers share:

$36 = \mathbf{2} \times \mathbf{2} \times \mathbf{3} \times 3$

$60 = \mathbf{2} \times \mathbf{2} \times \mathbf{3} \times 5$

Step 3 Multiply the shared primes: **2 × 2 × 3 = 12**. So **HCF = 12**.

Check: 36 ÷ 12 = 3 and 60 ÷ 12 = 5. The answers 3 and 5 have nothing in common, so no bigger number fits.

Use it: $\frac{36}{60} = \frac{36 \div 12}{60 \div 12} = \frac{3}{5}$

4 Lowest Common Multiple (LCM)

What does LCM mean?

The **smallest number that is in both times tables** (a multiple of both numbers). **Used for:** a common denominator when adding or comparing fractions.

Worked example 3: find the LCM of 36 and 60

Step 1 Break both numbers down, as before:

$36 = 2 \times 2 \times 3 \times 3 = 2^2 \times 3^2$

$60 = 2 \times 2 \times 3 \times 5 = 2^2 \times 3 \times 5$

Step 2 Mark the shared primes, and what is left over in 36 and left over in 60:

$36 = \mathbf{2} \times \mathbf{2} \times \mathbf{3} \times \mathbf{3}$

$60 = \mathbf{2} \times \mathbf{2} \times \mathbf{3} \times \mathbf{5}$

Step 3 Take the shared primes **once**, then everything that is left over:

$\mathbf{2 \times 2 \times 3} \times \mathbf{3} \times \mathbf{5} = \mathbf{180}$

So **LCM = 180**.

Check: 180 ÷ 36 = 5 and 180 ÷ 60 = 3. Both divide exactly. **Use it:** $\frac{1}{36}$ and $\frac{1}{60}$ can both be written with denominator 180.

Watch out!

HCF is never bigger than the smaller number. LCM is never smaller than the bigger number. For 36 and 60: HCF = 12 (small), LCM = 180 (big).

Do not count the shared primes twice in the LCM: 2 × 2 × 3 are taken **once**.

5 Summary

	HCF	LCM
Name	Highest Common Factor	Lowest Common Multiple
Meaning	The biggest number that divides into both	The smallest number that both divide into
Size	Small: at most the smaller number	Big: at least the bigger number
Method	Break both down. Multiply the primes they **share**.	Break both down. Shared primes **once**, times what is **left over** in each.
Used for	Simplifying fractions	Common denominators; problems of the type "when do they happen together again?"
Example	36 and 60: **12**	36 and 60: **180**

Exercise write the factorisation under every number you break down

1 Use a **division ladder** in your notebook. Write the factorisation here.

a 24 = _____

b 45 = _____

c 84 = _____

2 Use a **factor tree** in your notebook. Write the factorisation here.

a 72 = _____

b 90 = _____

3 Break both numbers down, then find the HCF and the LCM.

a 18 and 27 **b** 20 and 30 **c** 28 and 42 **d** 8 and 9

4 Tom says: *"The HCF of 8 and 12 is 24."* What has he found instead? Write the correct HCF.

5 **a** Leo has 24 red beads and 36 blue beads. He makes identical bracelets and uses every bead. What is the greatest number of bracelets he can make? How many beads of each colour are on one bracelet?
b Priya's two lights flash every 12 s and every 18 s. They flash together now. After how many seconds do they flash together again?

6 Simplify using the HCF: **a** $\frac{36}{60}$ **b** $\frac{28}{42}$

? What do you think?

1 The HCF of 7 and 5 is 1. Why? What is their LCM?

2 How can you tell, before you start, that the LCM of two numbers will be bigger than both?

CHALLENGE

Two numbers have HCF 4 and LCM 24. One of them is 8. Find the other.

Summary checklist

- ☐ I can break a number into prime factors with a ladder and a tree.
- ☐ I can find the HCF and say what it means.
- ☐ I can find the LCM and say what it means.